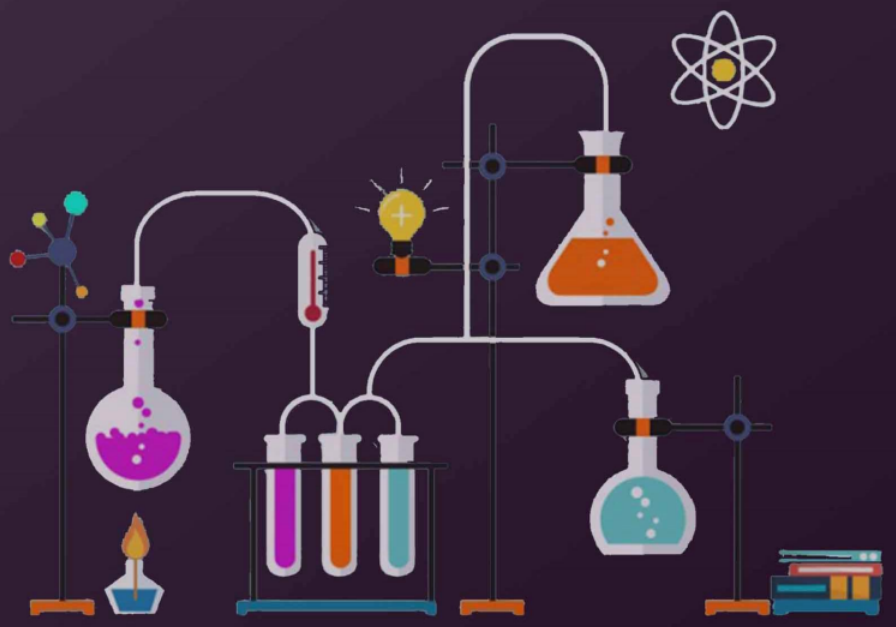
CHEMISTRY

# Thermodynamics



## SHEET

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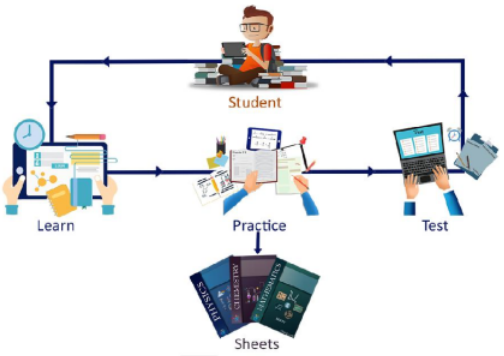
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***Note***

*Detailed solutions are available on the eSaral App.*

# *Module Description*




For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

*"Practice makes a man perfect"*

**1. Concept builder – 1 & 1A**

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

**2. Brain Booster – 2 & 2A**

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

**3. Simulator – JM & JA**

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

# *Home Work Index*

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

| Sr.No. | Topic Name | Ex 1 | Ex 1A | Ex 2 | Ex 2A |
|---|---|---|---|---|---|
| 1 | Introduction to Thermodynamics | 1,2 | 1, 19-21, 34 | 1-4 | 1,4 |
| 2 | First Law of Thermodynamics | 3,4 | 2 | | |
| 3 | Types of Processes and Work Done | 5-8 | 3-5,32,33 | 5 | 32,33 |
| 4 | Heat, Degrees of Freedom and Internal Energy | 9-13 | 38 | 6-8 | 2,3,36,37 |
| 5 | Enthalpy | 14-18 | 17,24, 35-37 | 9 | 10,19-21, 28,31 |
| 6 | Isothermal Process in Detail | 19-22 | 6,33 | | |
| 7 | Free Expansion, Adiabatic and Polytropic Process | 23-26 | 7-12,18,30 | 10 | 6,7,35 |
| 8 | Cyclic Process | | | 11-13 | |
| 9 | Carnot Cycle and Entropy | 27-30 | | 14,15 | 8,14 |
| 10 | Second Law of Thermodynamics | 31-34 | 14,22 | 16 | |
| 11 | Entropy Change in various processes | 35-40 | 13,23 | 17-21 | 11-13,38 |
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## Exercise - 1 Concept Builder

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

### Introduction to Thermodynamics

**Q1.** In a closed system.
(A) Energy is not exchanged
(B) Matter is exchanged
(C) Energy is only exchanged
(D) Energy and matter are exchanged

**Q2.** Which one of the following statement is false.
(A) Work is a state function
(B) Temperature is a state function
(C) Change in the state is completely defined when the initial and final states are specified
(D) Work appears at the boundary of the system

### First Law of Thermodynamics

**Q3.** In a process, 701 J of heat is absorbed by a system and 394 J of work is done by the system.
What is the change in internal energy for the process ?
(A) + 307 J (B) – 394 J
(C) + 701 J (D) – 307 J

**Q4.** According to 1 st law of Thermodynamics.
(A) The energy of system is constant
(B) The energy of universe is constant
(C) The energy of surroundings is constant
(D) The energy of system and surroundings are not constant

### Types of Processes and Work Done

**Q5.** If the internal energy of an ideal gas decreases by the same amount as the work done by the system, the process is :-
(A) cyclic (B) isothermal
(C) adiabatic (D) isolated

**Q6.** Which is an irreversible process ?
(A) Mixing of two gases by diffusion
(B) Evaporation of water at 373K and 1atm pressure
(C) Dissolution of NaCl in water
(D) Burning of Coal
The correct answer
(A) A,B (B) B,C
(C) A,B,C (D) A,C,D

**Q7.** The work done when a gas is compressed by an average pressure of 0.50 atm so as to decrease its volume from 400 $cm^3$ to 200$cm^3$
(A) 10.13 J (B) 20.13J
(C) 30.13 J (D) 40.13 J

**Q8.** In a closed insulated container, a liquid is stirred with a paddle to increase the temperature. Which of the the following is true ?
(A) $\Delta U = W \neq 0, q = 0$
(B) $\Delta U = W = 0, q \neq 0$
(C) $\Delta U = 0, W = q \neq 0$
(D) $W = 0, \Delta U = q \neq 0$

### Heat, Degree of Freedom and Internal Energy

**Q9.** One mole of a real gas is subjected to heating at constant volume from ($P_1$, $V_1$, $T_1$) state to ($P_2$, $V_1$, $T_2$) state. Then it is subjected to irreversible adiabatic compression against constant external pressure of $P_3$ atm till system reaches the final state ($P_3$, $V_2$, $T_3$).

If the constant volume molar heat capacity of real gas is $C_V$. Find out correct expression for $\Delta H$ from state 1 to state 3 -

(A) $C_v(T_3-T_1)+(P_3V_1-P_1V_1)$

(B) $C_v(T_2-T_1)+(P_3V_2-P_1V_1)$

(C) $C_v(T_2-T_1)+(P_3V_1-P_1V_1)$

(D) $C_p(T_2-T_1)+(P_3V_1-P_1V_1)$

**Q10.** One mole of an ideal monoatomic gas at temperature T and volume 1L expands to 2L against a constant external pressure of one atm under adiabatic conditions, then final temperature of gas will be :-

(A) $T+\frac{2}{3\times0.0821}$ (B) $T-\frac{2}{3\times0.0821}$

(C) $\frac{T}{2^{5/3-1}}$ (D) $\frac{T}{2^{5/3+1}}$

**Q11.** For a monatomic gas, the value of the ratio of $C_{p.m}$ and $C_{v.m}$ is :-

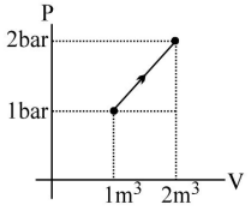
(A) $\frac{5}{3}$ (B) $\frac{7}{5}$ (C) $\frac{9}{7}$ (D) $\frac{9}{11}$

**Q12.** For a gas having molar mass M, specific heat at constant pressure can be given as :

(A) $\frac{\gamma R}{M(\gamma-1)}$ (B) $\frac{\gamma}{RM}$

(C) $\frac{M}{R(\gamma-1)}$ (D) $\frac{\gamma RM}{\gamma+1}$

**Q13.** What is $\Delta U$ for the process described by figure. Heat supplied during the process q = 100 kJ.




(A) + 50 kJ (B) – 50 kJ

(C) –150 kJ (D) + 150 kJ

## Enthalpy

**Q14.** Out of U, H, q, W and S which are state function?

(A) U, H, W (B) U, S, H, W

(C) U, H, S (D) U, H, q, W, S

**Q15.** Enthalpy of the system is given as :-

(A) H + PV (B) U + PV

(C) U – PV (D) H – PV

**Q16.** For the gas - phase decomposition,

$PCl_5(g) \overset{\Delta}{\rightleftharpoons} PCl_3(g) + Cl_2(g)$ :-

(A) $\Delta H<0, \Delta S<0$ (B) $\Delta H>0, \Delta S>0$

(C) $\Delta H>0, \Delta S<0$ (D) $\Delta H<0, \Delta S>0$

**Q17.** For the reaction,

$C_2H_4(g)+3O_2(g)\rightarrow 2CO_2(g)+2H_2O(l)$;

$\Delta E=-1415\,KJ.$

The $\Delta H$ at $27^0C$ is -

(A) -1410 KJ (B) -1420 KJ

(C) +1420 KJ (D) +1410 KJ

**Q18.** One mole of ideal gas expands freely at 310 K from five litre volume to 10 litre volume. Then $\Delta U$ and $\Delta H$ of the process are respectively.

(A) 0 and 5 cal (B) 0 and 5 × 300 cal

(C) 0 and 0 (D) 5 and 0 cal

## Isothermal Process In Detail

**Q19.** The incorrect expression among the following is :

(A) $\frac{\Delta G_{system}}{\Delta S_{Total}}=-T$

(B) In isothermal process,

$$w_{reversible}=-nRT\log_e\left(\frac{V_f}{V_i}\right)$$

(C) $\text{In } K=\frac{\Delta H^o-T\Delta S^o}{RT}$

(D) $K=e^{-\Delta G^o/RT}$

**Q20.** Find the work done when 2 mole of hydrogen expand isothermally from 15 to 50 litres against a constant pressure of 1 atm at 25°C.

(A) 847.0 cal (B) 847 K cal

(C) 84.7 cal (D) 84.7 K cal

**Q21.** Calculate the work done when 2 moles of hydrogen expand isothermally and reversibly at 27°C from 15 to 50 litres.

(A) 14.45 K cal (B) 1445 J

(C) 1445 cals (D) 14.45 KJ

**Q22.** A gas expands isothermally and reversibly. The work done by the gas is -

(A) Zero

(B) Minimum

(C) Maximum

(D) Cannot be determined

## Free Expansion, Adiabatic and Polytropic Process

**Q23.** Which of the following statement is incorrect when a system consisting of ideal gas undergoes change of state.

(A) In reversible change, $\Delta S_{univ.} = 0$

(B) In an irreversible change, $\Delta S_{univ.} \neq 0$

(C) In an adiabatic process always $\Delta S_{syst.} = 0$

(D) In an isochoric process, $\Delta S_{syst.} = nC_v \ell n\left(\frac{T_2}{T_1}\right)$

**Q24.** When a gas is subjected to adiabatic expansion it gets cooled due to -

(A) Loss in heat

(B) Increase in internal energy

(C) Due to repulsion between molecule

(D) Energy spent in doing work

**Q25.** $\Delta U = 0$ **for**

(A) Cyclic process, Adiabatic process

(B) Isothermal, Adiabatic process

(C) Cyclic process, Isothermal process

(D) Isochoric process, Isothermal process

**Q26.** One mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27°C. If the work done by the gas in the process is 3 kJ, the final temperature will be equal to ($C_V$=20 J/K mol)

(A) 100 K (B) 450 K

(C) 150 K (D) 400 K

## Carnot Cycle and Entropy

**Q27.** The maximum efficiency of a heat engine operating between 100°C and 25°C is

(A) 20.11% (B) 22.2%

(C) 25.17% (D) None

**Q28.** 9.0 gm of $H_2O$ is vaporised at 100°C and 1 atm pressure. If the latent heat of vaporisation of water is xJ/gm, then ΔS is given by :-

(A) $\frac{x}{373}$ (B) $\frac{18x}{100}$

(C) $\frac{18x}{373}$ (D) $\frac{1}{2} \times \frac{18x}{373}$

**Q29.** For conversion C(graphite) → C(Diamond), the ΔS is :-

(A) Zero (B) Positive

(C) Negative (D) Unknown

**Q30.** For isothermal expansion in case of an ideal gas:

(A) $\Delta G = \Delta S$

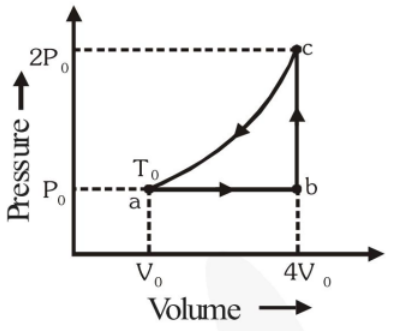
(B) $\Delta G = \Delta H$

(C) $\Delta G = V.\Delta P$

(D) $\Delta G = -V.\Delta P$

## Second Law of Thermodynamics

**Q31.** One mole of an ideal monoatomic gas is caused to go through the cycle shown in figure. Then the change in the internal energy in expending the gas from a to c along the path abc is :-




(A) $3P_0V_0$ (B) $6RT_0$
(C) $4.5RT_0$ (D) $10.5RT_0$

**Q32.** The correct relationship between free energy change in a reaction and the corresponding equilibrium constant $K_C$ is :-
(A) $\Delta G^\circ = RT\ \ell nK_C$
(B) $-\Delta G^\circ = RT\ \ell nK_C$
(C) $\Delta G = RT\ \ell nK_C$
(D) $-\Delta G = RT\ \ell nK_C$

**Q33.** Predict which of the following reaction(s) has a positive entropy change ?
(I) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
(II) $NH_4Cl(s) \rightarrow NH_3(g) + HCl(g)$
(III) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$
(A) I and II (B) III
(C) II and III (D) II

**Q34.** Among the following, the reaction for which $\Delta H = \Delta U$ is-
(A) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
(B) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$
(C) $C_2H_5OH(\ell) + 3O_2(g) \rightleftharpoons 2CO_2(g) + 3H_2O(\ell)$
(D) $C_2H_4(g) + H_2(g) \rightleftharpoons C_2H_6(g)$

## Entropy Change In Various Processes

**Q35.** One mole of an ideal diatomic gas ($C_V$ = 5 cal) was transformed from initial 25°C and 1L to the state when temperature is 100°C and volume 10L. The entropy change of the process can be expressed as (R = 2 calories /mol / k) :-
(A) $3\ell n\dfrac{298}{373} + 2\ell n\ 10$
(B) $5\ell n\dfrac{373}{298} + 2\ell n\ 10$
(C) $7\ell n\dfrac{373}{298} + 2\ell n\dfrac{1}{10}$
(D) $5\ell n\dfrac{373}{298} + 2\ell n\dfrac{1}{10}$

**Q36.** In the evaporation of water, the entropy :-
(A) decrease
(B) increase
(C) does not change
(D) sometimes increases, sometimes decreases

**Q37.** In which reactions $\Delta S$ is positive
(A) $H_2O(\ell) \longrightarrow H_2O(s)$
(B) $3O_2(g) \longrightarrow 2O_3(g)$
(C) $H_2O(\ell) \longrightarrow H_2O(g)$
(D) $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

**Q38.** Entropy change for an adiabatic reversible process is
(A) Positive (B) Zero
(C) Negative (D) Constant

**Q39.** For a reaction to be spontaneous at all temperatures :-
(A) $\Delta G$ – ve, $\Delta H$ + ve and $\Delta S$ + ve
(B) $\Delta G$ + ve, $\Delta H$ – ve and $\Delta S$ + ve
(C) $\Delta G$ – ve, $\Delta H$ – ve and $\Delta S$ – ve
(D) $\Delta G$ – ve, $\Delta H$ – ve and $\Delta S$ + ve

**Q40.** Zinc reacts with dilute hydrochloric acid to give hydrogen at 17°C. At this temperature, enthalpy change and entropy change are –12.55 KJ mol $^{-1}$ and 5.0 J $K^{-1}$ $mol^{-1}$ respectively. Then :-

(A) $\Delta G = -14$KJ $mol^{-1}$, reaction will be spontaneous

(B) $\Delta G = +14$ KJ $mol^{-1}$, reaction will be non spontaneous

(C) $\Delta G = -11.1$ KJ $mol^{-1}$, reaction will be spontaneous

(D) $\Delta G = +11.1$ KJ $mol^{-1}$, reaction will be non-spontaneous

## Third Law of Thermodynamics and T-S diagram

**Q41.** Identify the correct statement regarding entropy

(A) At 0°C, the entropy of a perfectly crystalline substance is taken to be zero

(B) At absolute zero of temperature, the entropy of a perfectly crystalline substance is +ve

(C) At absolute zero of temperature, the entropy of all crystalline substances is taken to be zero.

(D) At absolute zero of temperature, the entropy of a perfectly crystalline substance is taken to be zero.

**Q42.** A particular reaction at 27° C for which $\Delta H > 0$ and $\Delta S > 0$ is found to be non-spontaneous.

The reaction may be proceed spontaneously if :-

(A) The tempreture is decreased

(B) The tempreture kept constant

(C) Tempreture is increased

(D) It is carried in open vessel at 27°C

## Spontaneity and Gibb's Energy

**Q43.** For the change $H_2O(l) \rightarrow H_2O(g)$ ; P = 1 atm, T = 373 K, the free energy change $\Delta G = 0$. This indicates that :-

(A) $H_2O(l)$ is in equilibrium with $H_2O(g)$

(B) Water boils spontaneously at 373 K

(C) Water does not boil spontaneously at 373 K

(D) Condensation of water vapour occurs spontaneously at 373 K

**Q44.** Quick lime CaO is produced by heating lime stone $CaCO_3$ to drive of $CO_2$ gas

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ ;

$\Delta H° = 178.3$ kJ $\Delta S° = 160$ J/k

At what temperature the reaction will be equilibrium :-

(A) 1114 °C (B) 1468 K

(C) 1114 K (D) 841 K

**Q45.** $\Delta G°$ for the reaction X + Y ⇌ Z is –4.606 kcal. The value of equilibrium constant of the reaction at 227°C is ($R = 2.0$ cal. $mol^{-1}$ $K^{-1}$).

(A) 100 (B) 10 (C) 2 (D) 0.01

## Gibb's Energy In Detail

**Q46.** The standard free energy change ($\Delta G°$) is related to equilibrium constant (K) as :-

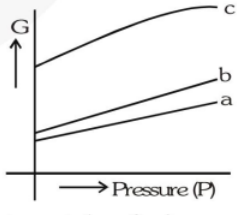
(A) $\Delta G° = -2.303$ RT log K

(B) $\Delta G° = 2.303$ RT log K

(C) $\Delta G° = RT \log K$

(D) $\Delta G° = -RT \log K$

**Q47.** The following curve represents the variation of Gibbs function 'G' with pressure at constant temperature.




Correct match of given plots with the physical state of a substance is :-

(A) c – solid, a – gas, b – liquid

(B) c – gas, b – liquid, a – solid

(C) a – liquid, b – solid, c – gas

(D) c – gas, b – solid, a – liquid

**Q48.** Standard Gibbs energy change $\Delta G^\circ$ for a reaction is zero. The value of the equilibrium constant will be :-

(A) 1 (B) 10

(C) 100 (D) $\infty$

**Q49.** The dependence of Gibbs free energy on pressure for an isothermal process of an ideal gas is given by :-

(A) $\Delta G_T = nRT \ln \frac{P_2}{P_1}$

(B) $\Delta G_T = nRT \ln \frac{V_2}{V_1}$

(C) $\Delta G_T = nRT \log \frac{P_2}{P_1}$

(D) $\Delta G_T = nRT \log \frac{V_2}{V_1}$

## ACC

**Q50.** In the process of ice melting at –15°C :-

(A) $\Delta G < 0$ (B) $\Delta G > 0$

(C) $\Delta G = 0$ (D) Cannot predict

**Q51.** For a reaction, $A + B \rightarrow AB$, $\Delta C_p$ is given by the equation $40 + 5 \times 10^{-3}$ T $JK^{-1}$ in the temperature range 300 – 600 K. The enthalpy of the reaction at 300 K is –25.0 kJ.

Calculate the enthalpy of the reaction at 450 K.

Also, $S^\circ (NO_2) = 57.5$ cal/deg,

$S^\circ(O_2) = 49.0$ cal/deg,

$S^\circ(NO) = 50.3$ cal/deg,

$S^\circ (O_3) = 56.8$ cal/deg.

(A) – 10.12 kJ (B) – 28.32 kJ

(C) – 18.72 kJ (D) – 8.21 kJ

**Q52.** Combustion of octane takes place in an automobile engine. The homogeneous equation of combustion is -

$$C_8H_{18}(g) + \frac{25}{2} O_2(g) \rightarrow 8CO_2(g) + 9H_2O(g)$$

The signs of $\Delta H$, $\Delta S$ and $\Delta G$ for the reaction will be :

(A) +ve, -ve, +ve (B) -ve, +ve, -ve

(C) -ve, +ve, +ve (D) +ve, +ve, -ve

# *Exercise - 1A* *Concept Builder*

## ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s).* ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** System that can exchange energy or matter with surrounding :-

(A) isolated system (B) closed system

(C) open system (D) none

**Q2.** If work done in adiabatic expansion is w & change in internal energy is $\Delta U$ then correct statement(s):-

(A) $w < 0, \Delta U < 0$ (B) $w > 0, \Delta U < 0$

(C) $w > 0, \Delta U > 0$ (D) $w = \Delta U$

**Q3.** Which of the following is / are example of non-PV work :-

(A) electric work

(B) mechanical work

(C) nerves activity

(D) muscular activity

**Q4.** From some initial state $(P_1, V_1, T_1)$ to $P_2$ two identical gas sample were expanded by

(I) Reversible adiabatic expansion

(II) Reversible isothermal expansion

then what is / are the correct option :-

(A) $w_I > w_{II}$

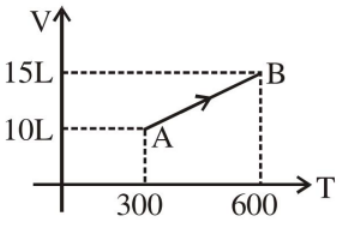
(B) $|w_I| < |w_{II}|$

(C) $(V_2)_{II} > (V_2)_I$

(D) $(T_2)_{II} > (T_2)_I$

**Q5.** If one mole monoatomic ideal gas was taken through process AB as shown in figure, then select correct option(s).

Given : ln 1.5 = 0.4




(A) $w_{AB} = -1496.52$ J

(B) $q_{AB} = 5237.82$ J

(C) $\Delta H_{AB} = 3741.3$ J

(D) $\Delta S_{AB}$ is + ve

**Q6.** In isothermal ideal gas compression :

(A) w is + ve (B) $\Delta H$ is zero

(C) $\Delta S_{gas}$ is + ve (D) $\Delta G$ is + ve

**Q7.** Which of the following is/are represent irreversible process :-

(A) mixing of two different gaseous substance

(B) water falling from the top

(C) combustion of paper

(D) friction less movement

**Q8.** For an isobaric process $\Delta H$ is represented by or equal to :-

(A) w (B) $\Delta U$

(C) q (D) none

**Q9.** Which of the following is / are correct :-

(A) heat involve in isochoric process for an ideal gas is equal to change in internal energy.

(B) for an ideal gas in a isothermal process work done by the system is equal to heat absorbed by the system.

(C) heat involve in adiabatic process for an ideal gas is zero

(D) heat involve in isobaric process for an ideal gas is equal to change in enthalpy

**Q10.** An adiabatic expansion is one in which :-

(A) all energy is transferred as heat

(B) no energy is transferred as heat

(C) the temperature of a gas decreases in a reversible adiabatic expansion

(D) $du \neq dw$

**Q11.** For free expansion process :-

(A) $\Delta S_{system} = 0$ (B) $\Delta S_{surr.} = 0$

(C) $\Delta S_{net} > 0$ (D) $\Delta S_{net} = 0$

**Q.12** Which of the following is true for reversible adiabatic process involving an ideal gas?

(A) Gas with higher $\gamma$ has high magnitude of slope in a P (y-axis) v/s T (x-axis) curve

(B) Gas with higher $\gamma$ has high magnitude of slope in a V (y-axis) v/s T (x-axis) curve

(C) Gas with higher $\gamma$ has high magnitude of slope in a P (y-axis) v/s V (x-axis) curve

(D) Gas with higher $\gamma$ has low magnitude of slope in a P (y-axis) v/s T (x-axis) curve

**Q13.** Select the correct statements for the equilibrium under standard condition :-

$H_2O(s) \rightleftharpoons H_2O(\ell); \Delta S_1^\circ$

$H_2O(\ell) \rightleftharpoons H_2O(g); \Delta S_2^\circ$

$H_2O(s) \rightleftharpoons H_2O(g); \Delta S_3^\circ$

(A) $\Delta S_1^\circ > \Delta S_2^\circ$ (B) $\Delta S_2^\circ >>> \Delta S_1^\circ$

(C) $\Delta S_3^\circ > \Delta S_2^\circ$ (D) $\Delta S_3^\circ >>> \Delta S_1^\circ$

**Q14.** Which of the following statement (s) is/are correct ?

(A) The system of constant entropy and constant volume will attain the equilibrium in a state of minimum energy.

(B) The entropy of the universe is increasing.

(C) The process would be spontaneous when $(\Delta S)_{E,V} < 0$ and $(\Delta E)_{S,V} > 0$

(D) The process would be spontaneous when $(\Delta S)_{E,V} > 0$ and $(\Delta E)_{S,V} < 0$

**Q15.** Which of the following is/are correct ?

(A) $\Delta G = \Delta H - T\Delta S$

(B) At equilibrium $\Delta G \neq 0$

(C) At equilibrium $\Delta G^\circ = -RT \ln K_{eq}$

(D) $\Delta G = \Delta G^\circ + RT \ln K_{eq}$

**Q16.** Which is / are the correct :-

(A) $\Delta_r G = \Delta_r G^\circ + RT\ell nQ^\circ$

(B) $\Delta_r G^\circ = \Delta_r G + RT\ell nQ^\circ$

(C) $\Delta_r G^\circ = -RT\ell nK^\circ_{aq}$

(D) $\Delta_r G^\circ = RT\ell nK^\circ_{aq}$

**Q17.** The bond dissociation enthalpies of $H_2(g)$ and $N_2(g)$ are + 435.95 k $mol^{-1}$ and + 941.8 kJ $mol^{-1}$ and enthalpy of formation of $NH_3(g)$ is – 46.024 kJ $mol^{-1}$. Choose the correct options -

(A) The enthalpy of atomization of $NH_3(g)$ is 1.170849 MJ $mol^{-1}$

(B) The bond enthalpy of N – H bond is 390.283 KJ $mol^{-1}$

(C) The bond enthalpy of N – H bond is 390.283 MJ $mol^{-1}$

(D) The enthalpy of atomization of $NH_3(g)$ is 1.170849 KJ $mol^{-1}$

**Q18.** Which statement is/are correct :-

(A) Final temperature in reversible adiabatic expansion is lesser than in irreversible adiabatic expansion.

(B) When heat is supplied to an ideal gas in an isothermal process, kinetic energy of gas will increase.

(C) When an ideal gas is subjected to adiabatic expansion it gets cooled

(D) Entropy increases in atomisation of dihydrogen

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions.Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

### Paragraph for Questions 19 to 21

Physical properties of materials and system can after be categorized as being either intensive or extensive quantities according to how the property changes when the size of system changes.

Intensive property is one whose magnitude is independent of the size OR mass of the system. An extensive property is one whose magnitude depend on the mass of system.

**Q19.** Which of the following is not a intensive prop. :-

(A) heat capacity (B) refractive index

(C) sp. volume (D) entropy

**Q20.** Select an extensive prop. :-

(A) melting point (B) molality

(C) pressure (D) internal energy

**Q21.** Correct for enthalpy :-

(A) intensive prop.

(B) extensive

(C) independent of mass of sys.

(D) none

### Paragraph for Questions 22 to 24

The change in Gibbs free energy ($\Delta G$) of the system alone provides a criterion for the spontaneity of a process at constant temperature and pressure. The change in free energy of a system at constant temperature and pressure will be

$\Delta G_{system} = \Delta H_{system} - T\Delta_{system}$

If $\Delta G$ = (–) ve process will be spontaneous

$\Delta G$ = (+) ve process will be non-spontaneous

$\Delta G = 0$ system at equilibrium

Standard free energy is related to the equilibrium constant as $\Delta G^\circ = -2.303\ RT \log K_{eq}$.

**Q22.** A process is spontaneous at all temperatures if –

(A) $\Delta H > 0$ and $\Delta S > 0$

(B) $\Delta H > 0$ and $\Delta S < 0$

(C) $\Delta H = 0$ and $\Delta S < 0$

(D) $\Delta H < 0$ and $\Delta S > 0$

**Q23.** What will be the entropy change when two moles of an ideal gas expands reversibly from initial volume of 1 litre to 10 litre at constant temperature 300 K ?

(A) 19.5 $JK^{-1}mol^{-1}$ (B) 38.27 $JK^{-1}mol^{-1}$

(C) 11.48 $kJK^{-1}mol^{-1}$ (D) 5.74 $kJK^{-1}mol^{-1}$

**Q24.** The enthalpy of vapourisation of a liquid is 30 kJ/mol and entropy of vapourisation is 75 J $mol^{-1}K^{-1}$. The boiling point of the liquid at 1 atm is -

(A) 250 K (B) 400 K

(C) 450 K (D) 600 K

**Paragraph for Questions 25 to 27**

Spontaneity of any process can be predicted with the help of $\Delta S_{total}$. But this requires calculation of changes in system as well as surroundings. If some criteria (depending upon the system only) can be developed for checking spontanity under specific conditions, then that would be a more useful parameter. The criteria can be derived from Clausius inequality.

$TdS > q$, > sign for reversible process = sign for reversible process

or $TdS > dU - W$ for an irreversible Process

or $TdS_{sys} > dU_{sys} + p\,dV$ [consider no non-PV work]

If V and T are costant

$TdS > dU$

or $dU - T\,dS < 0$

As temperature is constant, $dU - d(TS) < 0$

or $d(U - TS) < 0$

Another state function A(Helmholtz's function)

$= U - TS$

A decrease in Helmholts function (H) under constant volume and tempearture is the criteria of spontaneity of a process.

**Q25.** For a spontaneous process, if entropy and volume are constant, the internal energy of system must :-

(A) increase (B) decrease

(C) remain constant (D) be zero

**Q26.** If a state function G is defined such as $G = H - TS$, then dG must decrease for a spontaneous process (involving only PV work) occuring at :-

(A) Constant volume and temperature

(B) Constant pressure and temperature

(C) Constant volume and pressure

(D) Constant entropy and volume

**Q27.** For an irreversible cyclic process is $\oint \frac{dq}{T}$ is :-

(A) equal to zero

(B) greater than zero

(C) less than zero

(D) equal to change in Gibb's energy

## MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

**Q28.** Match the column-I with Column-II:-

| | **List - I** | **List - II** |
|---|---|---|
| (a) | Perfectly (crystalline solid) | (p) $\Delta U = 0$ |
| (b) | Reversible reaction at equilibrium | (q) T = constant |
| (c) | Isothermal process | (r) $\lim_{T\to 0K} S \to 0$ |
| (d) | $\left(\frac{\partial G}{\partial P}\right) = -S$ | (s) $\Delta S_{universe} = 0$ |

(A) a – r ; b – s ; c – p,q ; d – q (B) a – s ; b – r ; c – p,q ; d – q

(C) a – r ; b – p,q ; c – s ; d – q (D) a – q ; b – s ; c – p,q ; d – r

**Q29.** **List - I** **List - II**

(A) Process occurs under constant temperature (P) Adiabatic process

(B) Process occurs under no exchange of heat with the surroundings (Q) Isothermal process

(C) Cyclic integral of thermodynamic function is zero (R) Isochoric process

(D) Process occurs under constant volume condition (S) Reversible process.

| | A | B | C | D |
|---|---|---|---|---|
| (A) | P | R | S | Q |
| (B) | Q | P | S | R |
| (C) | Q | S | R | P |
| (D) | P | Q | R | S |

**Q30.** **List - I** **List - II**

(A) Isothermal process (P) $q = \Delta U$

(B) Adiabatic process (Q) $w = -P\Delta V$

(C) Isobaric process (R) $w = \Delta U$

(D) Isochoric process (S) $w = -nRT \ln (V_2/V_1)$

| | A | B | C | D |
|---|---|---|---|---|
| (A) | R | S | Q | P |
| (B) | Q | R | S | Q |
| (C) | Q | S | R | P |
| (D) | S | R | Q | P |

**Q31.** Match the physical changes in List-I with their relations given in List-II :-

**List-I** **List-II**

(A) $\Delta G$ (i) $\Delta U + P\Delta V$

(B) $\Delta H$ (ii) $-nEF$

(C) $\Delta S^o$ (iii) $-RT \log_e K$

(D) $\Delta G^o$ (iv) $nR \log_e \left(\frac{V_2}{V_1}\right)$

Select the correct answer form the given codes :

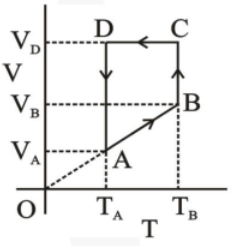
**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | (ii) | (i) | (iv) | (iii) |
| (B) | (i) | (ii) | (iii) | (iv) |
| (C) | (iv) | (iii) | (ii) | (i) |
| (D) | (i) | (ii) | (iv) | (iii) |

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q32.** A monoatomic ideal gas of two moles is taken through a cyclic process starting from A as shown in figure. The volume ratio are $\frac{V_B}{V_A} = 2$ and $\frac{V_D}{V_A} = 4$. If the temperature $T_A$ at A is 27 °C, then match the following.




| Column - I | Column- II |
|---|---|
| (A) Ratio of temperature at point B to point A | (P) – 1200 cal |
| (B) Ratio of heat exchange in process B → C to process D → A | (Q) – 1800 cal |
| (C) Work done in complete cyclic process | (R) 2 |
| (D) Heat exchange in process C → D | (S) – 1 |

**Q33.**

| Column-I (Process of reaction) | Column-II (Positive, negative) |
|---|---|
| (A) For the process $H_2O_{(\ell)} \rightleftharpoons H_2O_{(s)}$, ΔH & ΔS are | (P) –ve, +ve |
| (B) For the endothermic reaction $2A_{(s)} + \frac{1}{2}O_{2(g)} \rightleftharpoons A_2O_{(s)}$ at 298 K, ΔS & ΔG are | (Q) +ve, –ve |
| (C) C(diamond) ⇌ C(graphite), favourable conditions for formation of diamond are high pressure and high temperature then $\Delta H_f$ of diamond and ΔS of Diamond from Graphite are | (R) +ve, +ve |
| (D) For the given reaction $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$, $E_{a(forward)}$ = 57.2 kJ and $E_{a(backward)}$ = 3.2 kJ, ΔH & ΔS for the given reaction | (S) –ve, –ve |

## NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

**Q34.** If x & y are extensive properties, then number of intensive properties in the following :-

(1) $x + y$ ; (2) $x - y$ (3) $\frac{x}{y}$

(4) dx ; (5) $\frac{dx}{dy}$

**Q35.** 180 gm of water is evaporated slowly under isothermal conditions at 100°C. Assuming water vapour to behave ideally ($\Delta H_{vap.} = 2260$ J/g). What is the value of $\Delta S$: (Give answer in sum of all digit, untill it becomes between 0 to 9)

**Q36.** When 10 litre of ozone gas at 1 atm is compressed adiabatically to $1/5^{th}$ of its volume, the pressure becomes 6.5 times of the initial. Find out the work done in the process. (Assuming ideal behaviour for ozone, give your answer in L-atm)

**Q37.** If (at constant pressure) following reaction (1) takes place & heat evolve in this process is x & heat evolve in reaction (2) is y.

$$SO_2 + \frac{1}{2}O_2(g) \rightarrow SO_3(g) \ ; q = x \text{ kJ} \quad ...(1)$$

$$2SO_2 + O_2(g) \rightarrow 2SO_3(g) \ ; q = y \text{ kJ} \quad ...(2)$$

What is the value of $\frac{y}{x}$ ?

## ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

**Q38.** **Statement 1 :** Internal energy of a system is an extensive property.

**Statement 2 :** The internal energy of a system depends upon the amount and physical state of the substance.

# *Exercise - 2* *Brain Booster*

## SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

**Q1.** In thermodynamics, a process is called reversible when :

(A) surroundings and system change into each other

(B) there is no boundary between system and surroundings

(C) the surroundings are always in equilibrium with the system

(D) the system changes into the surroundings spontaneously

**Q2.** Which one of the following quantity is dependent on path ?

(A) molar internal energy

(B) volume

(C) w

(D) q + w

**Q3.** Out of boiling point (I), entropy (II), pH (III) and density (IV), Intensive properties are:

(A) I, II

(B) I,II, III

(C) I, III, IV

(D) All of these

**Q4.** In which of the following sets, all properties belong to same category (all extensive or all intensive)?

(A) Mass, Volume, Specific heat

(B) Temperature, Concentration, Volume

(C) Heat capacity, Concentration, Entropy

(D) Enthalpy, Entropy, Volume

**Q5.** Five moles of ideal gas expand isothermally and reversibly from an initial pressure of 100 atm to a final pressure of 1 atm at 27°C. The work done by the gas is -

(A) 3455 cal (B) 13818 cal

(C) 0 (D) 69010 cal

**Q6.** Internal energy does not include

(A) vibrational energy

(B) rotational energy

(C) nuclear energy

(D) energy arising by gravitational pull

**Q7.** Heat capacity ($C_V$) of an ideal gas is X KJ/mole/K. To rise its temperature from 298K to 318K, heat to be supplied per 10g gas will be (in KJ) [MW=16]

(A) 16X (B) 6.25X

(C) 32X (D) 12.5X

**Q8.** An adiabatic process is one in which there is no transfer of heat across the boundary between system and surroundings. For such a process.

(A) $P_{ext}\Delta V = 0$ (B) $q = w$

(C) $\Delta U = w$ (D) $\Delta U = 0$.

**Q9.** The enthalpy of fusion of ice is 6.025 kJ-mol$^{-1}$. What is the entropy change in the conversion of 1g of ice to water at 273 K and one atmospheric pressure :-

(A) 1.22 J-K$^{-1}$g$^{-1}$

(B) 2.63 J-K$^{-1}$-mol$^{-1}$

(C) 3.74 J-K$^{-1}$-mol$^{-1}$

(D) 4.53 J-K$^{-1}$-mol$^{-1}$

**Q10.** An ideal gas can be expanded from an initial state to a certain volume through two different processes, (A) $PV^2 = K$ and (B) $P = KV^2$, where K is a positive constant. Then, choose the correct option from the following.

(A) Final temperature in (A) will be greater than in (B)

(B) Final temperature in (B) will be greater than in (A)

(C) Work done by the gas in both the processes would be equal

(D) Total heat given to the gas in (A) is greater than in (B)

**Q11.** A gaseous system changes from state A($P_1$,$V_1$,$T_1$) to B ($P_2$,$V_2$,$T_2$), B to C ($P_3$,$V_3$$T_3$) and finally from C to A. The whole process may be called :

(A) Cyclic process

(B) Reversible process

(C) Isobaric process

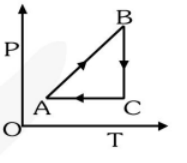
(D) Spontaneous process

**Q12.** The internal energy change when a system goes from state A to B is 40 KJ/mol. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the change in internal energy?

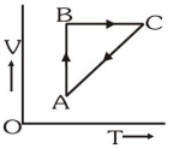
(A) Zero (B) 40KJ

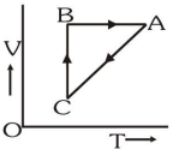
(C) >40 KJ (D) <40KJ

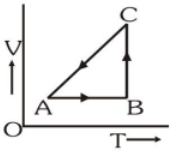
**Q13.** A cyclic process is shown in the P-T diagram.

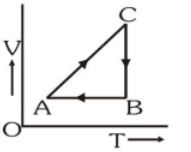



Which of the curves show the same process on V-T diagram ?

(A)



(B)



(C)



(D)


**Q14.** When two mole of an ideal gas $\left(C_{p,m}=\frac{5}{2}R\right)$ heated from 300 K to 600 K at constant pressure.
The change in entorpy of gass ($\Delta S$) is :-
(A) $\frac{3}{2}R \ln 2$ (B) $-\frac{3}{2}R \ln 2$
(C) $5R \ln 2$ (D) $\frac{5}{2}R \ln 2$

**Q15.** When heat is supplied to an ideal gas in an isothermal process, the -
(A) Gas will do positive work
(B) Gas will do negative work
(C) Kinetic energy of gas will increase
(D) Gas will not obey the law of conservation of energy

**Q16.** Considering entropy (S) as a thermodynamic parameter, the criterion for the spontaneity of any process is :-
(A) $\Delta S_{system} > 0$ only
(B) $\Delta S_{surroundings} > 0$ only
(C) $\Delta S_{system} + \Delta S_{surroundings} > 0$
(D) $\Delta S_{system} - \Delta S_{surroundings} > 0$

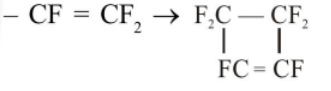
**Q17.** One mole of an ideal gas is allowed to expand isothermally and reversibly at 300 K until the volume has tripelled. The $\Delta S$ for the system is (log3 = 0.4771) :-
(A) 91.35 $Jk^{-1}$ (B) –91.35 $Jk^{-1}$
(C) zero (D) 9.135 $Jk^{-1}$

**Q18.** When two gases are mixed the entropy :–
(A) Remains constant (B) Decreases
(C) Increases (D) Becomes zero

**Q19.** Which of the following process proceed with increase of entropy :
I. Solid melts to liquid
II. Mixing of gases
III. Compression of gas
IV. Vaporisation of water
(A) I, II, III and IV (B) I, II and IV
(C) I and II only (D) II, III and IV

**Q20.** What is the entropy change for the conversion of 1gm ice to water at 273 K, $\Delta H_f$ for ice = 6.025 kJ $mol^{-1}$.
(A) 1.227 J $k^{-1}$ $gm^{-1}$
(B) 22.1 J $k^{-1}$ $mol^{-1}$
(C) 397.8 J $k^{-1}$ $gm^{-1}$
(D) 22.1 kJ $k^{-1}$ $mol^{-1}$

**Q21.** If enthalpy of vaporisation of water is 186.5 kJ/mol, the entropy of its vaporisation ($KJmol^{-1}K^{-1}$) will be :–
(A) 0.5 (B) 1.0
(C) 1.5 (D) 2.0

**Q22.** $F_2C = CF - CF = CF_2 \rightarrow$ 
For this reaction (ring closure) $\Delta H = -49$ kJ $mol^{-1}$ and $\Delta S = -40.2$ $JK^{-1}mol^{-1}$. upto what temperature is the forward reaction spontaneous ?
(A) 1492°C (B) 1219°C
(C) 946°C (D) 1080°C

**Q23.** The entropy of vaporization of benzene is 85 $JK^{-1}$ $mol^{-1}$. When 117g benzene vaporizes at it's normal boiling point, the entropy change of surrounding is :-
(A) – 85 $JK^{-1}$ (B) – 85 × 1.5 $JK^{-1}$
(C) 85 × 1.5 $JK^{-1}$ (D) None

**Q24.** Which of the following conditions are applied for conversion ice into water :-

| | $\Delta H$ | $\Delta S$ | $\Delta G$ |
|---|---|---|---|
| (A) | Negative | Negative | Negative at low T |
| (B) | Negative | Negative | Positive at low T |
| (C) | Positive | Positive | Positive at low T |
| (D) | Positive | Positive | Negative at low T |

**Q25.** The enthalpy and entropy change for a chemical reaction are $-2.5 \times 10^{-3}$ cal and 7.4 cal $deg^{-1}$ respectively. Predict whether the nature of reaction at 298 K is :-
(A) spontaneous (B) reversible
(C) irreversible (D) non-spontaneous

**Q26.** The favourable conditions for a spontaneous reaction are :-
(A) $T\Delta S > \Delta H$, $\Delta H$ = + ve, $\Delta S$ = + ve
(B) $|T\Delta S| < |\Delta H|$, $\Delta H$ = – ve, $\Delta S$ = – ve
(C) $|T\Delta S| = |\Delta H|$, $\Delta H$ = – ve, $\Delta S$ = + ve
(D) All of these

**Q27.** A reaction has $\Delta H = -33$ kJ and $\Delta S = -58$J/K. This reaction would be ?
(A) Spontaneous at all temperature
(B) Non-spontaneous at all temperature
(C) Spontaneous above a certain temperature only
(D) Spontaneous below a certain temperature only

**Q28.** Given the following data :-

| Substance | ΔH (kJ/mol) | ΔG (kJ/mol) |
|---|---|---|
| FeO(s) | –266.3 | –245.12 |
| C (Graphite) | 0 | 0 |
| Fe(s) | 0 | 0 |
| CO(g) | –110.5 | –137.15 |

Determine at what temperature the following reaction is spontaneous ?
$FeO(s) + C\ (Graphite) \rightarrow Fe(s) + CO(g)$
(A) 298 K
(B) 668 K
(C) 966 K
(D) $\Delta G$ is +ve, hence the reaction will never be spontaneous.

**Q29.** For the auto-ionization of water at 25°C, $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$ is $10^{-14}$.
What is $\Delta G°$ for the process ?
(A) $\approx 8\times10^4$ J (B) $\approx 3.5 \times 10^4$ J
(C) $\approx 10^4$ J (D) None of these

**Q30.** The value of $\log_{10}K$ for a reaction $A \rightleftharpoons B$ is :-
(Given : $\Delta_r H°_{298\ K} = -54.07$ kJ mol$^{-1}$,
$\Delta_r S°_{298\ K} = +10$ JK$^{-1}$ mol$^{-1}$
and $R = 8.314$ JK$^{-1}$ mol$^{-1}$;
$2.303 \times 8.314 \times 298 = 5705$) -
(A) 5 (B) 10
(C) 95 (D) 100

**Q31.** Heat of hydrogenation of ethene is $x_1$ and that of benzene is $x_2$. Hence resonance energy is-
(A) $x_1 - x_2$ (B) $x_1 + x_2$
(C) $3x_1 - x_2$ (D) $x_1 - 3x_2$

**Q32.** 44.0 kJ of heat is required to evaporate one mole of water at 298 K. If $\Delta H°_f$ of $H_2O(\ell)$ is –286 kJ mol$^{-1}$, $\Delta H°_f$ of $H_2O(g)$ is -
(A) – 330 kJ mol$^{-1}$ (B) + 242 kJ mol$^{-1}$
(C) –242 kJ mol$^{-1}$ (D) –198 kJ mol$^{-1}$

## *Exercise - 2A* — *Brain Booster*

### ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** Wall that can be used for "open system" :-
(A) diathermic
(B) permiable
(C) adiabatic
(D) rigid

**Q2.** For heat capacity which option is / are correct :-
(A) $C_P - C_V = R$ (B) $\frac{C_P}{C_V} = \gamma$
(C) $C_V = \left(\frac{\partial U}{\partial T}\right)_V$ (D) $C_P = \left(\frac{\partial H}{\partial T}\right)_P$

**Q3.** x : y is the ratio of $\gamma$ for $CH_4$ at high & low temperature then what is the value of $\frac{x}{y}$ :-
(A) $\frac{13}{16}$ (B) $\frac{13}{12}$ (C) $\frac{16}{13}$ (D) $\frac{4}{3}$

**Q4.** Which option(s) is/are represent state function :-

(A) $\int_1^2 dx = x_2 - x_1$

(B) $\oint dx = 0$ (cyclic integral)

(C) if F = f(x, y) then,

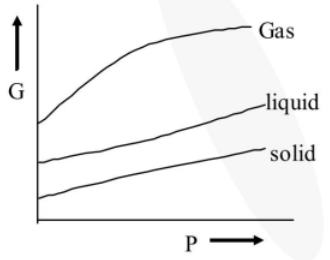
$$dF = \left(\frac{\partial F}{\partial x}\right)_y dx + \left(\frac{\partial F}{\partial y}\right)_x dy$$

(D) if F = f(x, y) then,

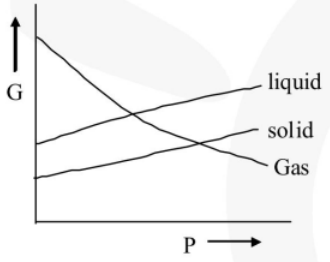
$$\frac{\partial}{\partial y}\left[\left(\frac{\partial F}{\partial x}\right)_y\right]_x = \frac{\partial}{\partial x}\left[\left(\frac{\partial F}{\partial y}\right)_x\right]_y$$

**Q5.** Which of the following correctly represent the variation of G.

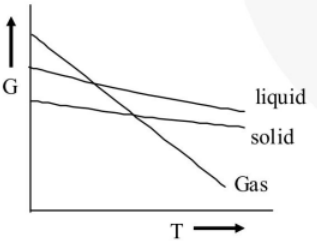
(A)



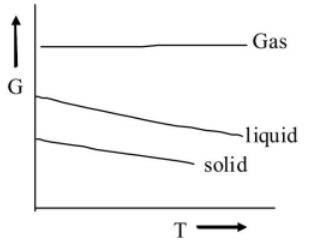

(B)




(C)




(D)




**Q6.** Find the incorrect option for PV-work :-

(A) area under PV-graph

(B) area under PT-graph

(C) area under VT-graph

(D) area under P, 1/V-graph

**Q7.** The work involved in a reversible adiabatic process of expansion of an ideal gas from $P_1$ and $V_1$ to $P_2$ and $V_2$ is given by-

(A) $W = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$

(B) $W = -nC_VT_1\left[1 - \left(\frac{P_2}{P_1}\right)^{R/C_P}\right]$

(C) $W = nCv\ (T_2 - T_1)$

(D) W = [efficiency of carnot cycle × $q_{absorbed}$]

**Q8.** The efficiency of a heat engine is maximum when :-

(A) temperature of source is greater than that of sink.

(B) temperature of sink is greater than that of source

(C) temperature difference of source and sink is minimum.

(D) temperature difference of source and sink is maximum

**Q9.** In which of the following reaction $\Delta_rS^\circ > 0$

(A) $SO_2(g) + \frac{1}{2}O_2(g) \longrightarrow SO_3(g)$

(B) $2O_3(g) \longrightarrow 3O_2(g)$

(C) $NaCl(s) \longrightarrow NaCl(aq)$

(D) $H_2(g) \longrightarrow 2H(g)$

**Q10.** For which of the following :-

$\Delta_rH > \Delta_rU$ (at constant T) (assume all gase are ideal)

(A) $CH_4(g) + 2O_2(g) \rightleftharpoons CO_2(g) + 2H_2O(\ell)$

(B) $2NO_2(g) \rightleftharpoons N_2(g) + 2O_2(g)$

(C) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(D) none

**Q11.** Which option is / are correct :-

(A) if $\Delta S_{sys.} + \Delta S_{surr.} > 0$ then process is spontaneous

(B) if $\Delta S_{sys.} + \Delta S_{surr.} = 0$ then process is reversible

(C) if $\Delta S_{sys.} + \Delta S_{surr.} < 0$ then process is non-spontaneous

(D) at equilibrium state $\Delta S_{sys.} + \Delta S_{surr.} = 0$

**Q12.** Which statements are correct ?

(A) A spontaneous chemical reaction which starts far from equilibrium always gives irreversible process.

(B) The entropy increases in an irreversible adiabatic process.

(C) The entropy decreases in a reversible adiabatic process.

(D) The entropy does not change in a reversible adiabatic process.

**Q13.** Which of the following statement (s) is/are false :

(A) $\Delta_r S$ for $\frac{1}{2}N_2(g) \longrightarrow N(g)$ is positive

(B) $\Delta G_{system}$ is always zero for a reversible process in a closed system

(C) $\Delta G^\circ$ for an ideal gas is a function of temperature and pressure

(D) Entropy of a closed system is always maximized at equilibrium

**Q14.** The efficiency of the Carnot Engine is 1/6. On decreasing the temperature of the sink by 65k, the efficiency increases to 1/3. Find the temperature of the source ?

(A) 325 k (B) 390 k

(C) 230 k (D) None

**Q15.** Identify the incorrect statement regarding sponteneous process

(A) For a spontaneous process in an isolated system, the change in entropy is positive

(B) Endothermic processes are never spontaneous

(C) Exothermic processes are always spontaneous

(D) Lowering of energy and Higher of entropy in the reaction process is the criterion for spontaneity

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph For Questions 16 to 18**

The coefficients thermal expansion $\alpha$ and the coefficient of compressibility, $\beta$ are defined as follows :

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P \text{ and } \beta = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$$

For a real gas (x), $\alpha = \frac{(V-a)}{TV}$ and $\beta = \frac{3(V-a)}{4PV}$

Where a is constant.

**Q16.** For one mole of an ideal gas $\alpha$ is

(A) $\frac{1}{V}$ (B) $\frac{1}{T}$

(C) $\frac{1}{TV}$ (D) $\frac{1}{P}$

**Q17.** For one mole of vanderwaals gas having negligible intermolecular attraction among molecules which of the following is/are correct?

(A) $\beta$ is directly proportional to compressibility factor, Z

(B) Z is directly proportional to $\beta$

(C) $Z = -\frac{P}{V}\left(\frac{\partial P}{\partial V}\right)_T$

(D) $Z = -\frac{V}{P}\left(\frac{\partial P}{\partial V}\right)_T$

**Q18.** What is the equation of state of real gas (x)? A represent a constant

(A) $P(V-a) = AT$ (B) $P(V-a)^{3/4} = AT$

(C) $P^{3/4}(V-a) = AT$ (D) $\frac{P^{3/4}}{(V-a)} = AT$

**Paragraph For Questions 19 to 21**

**Enthalpy of reaction :** It is used for the heat change accompanying any reaction depending upon the nature of the reaction the enthalpy of the reaction is named accordingly

For example : In the reaction

$$CH_4(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(\ell)\ ;$$
$$\Delta H = -890.4 \text{ kJ/mole}$$

This reaction shown that 890.4 kJ of heat is produced when methane is completely burnt it is known as heat of combustion.

For example: In the reaction

$$C(s) + O_2(g) \longrightarrow CO_2(g);$$
$$\Delta H = -393.5 \text{ kJ/mole}$$

when $CO_2(g)$ is formed from its elements i.e. from C(s) and $O_2(g)$ 393.5 kJ of heat is produced and the process in heat of formation.

**Q19.** The enthalpy of formation of methane from the following data

(i) $C(s) + O_2(g) \longrightarrow CO_2(g)$
$\Delta H = -393.7$ kJ/mole

(ii) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$
$\Delta H = -285.8$ kJ/mole

(iii) $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$
$\Delta H = -890.4$ kJ/mol

(A) –74.9 kJ/mole (B) 54.6 kJ/mole
(C) 99.7 kJ/mole (D) –89.6 kJ/mole

**Q20.** Calculate the enthalpy of formation of sucrose ($C_{12}H_{22}O_{11}$) from the following data:

(i) $C_{12}H_{22}O_{11} + 12O_2(g) \longrightarrow 12CO_2(g) + 11H_2O(l)$,
$\Delta H = -5200.7$ kJ mol$^{-1}$

(ii) $C(s) + O_2(g) \longrightarrow CO_2(g)$,
$\Delta H = -394.5$ kJ mol$^{-1}$

(iii) $H_2(g) + (1/2)\ O_2(g) \longrightarrow H_2O(l)$,
$\Delta H = -285.8$ kJ mol$^{-1}$

(A) –10.3 kJ/mole
(B) + 863.4 kJ/mole
(C) –2677.1 kJ/mole
(D) –4321.3 kJ/mole

**Q21.** Calculate the enthalpy of combustion of benzene from the following data:

(i) $6C(s) + 3H_2(g) \longrightarrow C_6H_6(l)$,
$\Delta H = 49.0$ kJ mol$^{-1}$

(ii) $H_2(g) + (1/2)\ O_2(g) \longrightarrow H_2O(l)$,
$\Delta H = -285.8$ kJ mol$^{-1}$

(iii) $C(s) + O_2(g) \longrightarrow CO_2(g)$,
$\Delta H = -389.3$ kJ mol$^{-1}$

(A) + 463.6 kJ/mole
(B) +32.4 kJ/mole
(C) – 1432.3 kJ/mole
(D) –3242.2 kJ/mole

**Paragraph For Questions 22 to 26**

Standard Gibb's energy of reaction ($\Delta_r G°$) at a certain temperature can be computed as $\Delta_r G° = \Delta_r H° - T.\Delta_r S°$ and the change in the value of $\Delta_r H°$ and $\Delta_r S°$ for a reaction with temperature can be computed as follows :

$$\Delta_r H°_{T_2} - \Delta_r H°_{T_1} = \Delta_r C°_p (T_2 - T_1)$$

$$\Delta_r S°_{T_2} - \Delta_r S°_{T_1} = \Delta_r C°_p \ln\left(\frac{T_2}{T_1}\right)$$

$$\Delta_r G° = \Delta_r H° - T.\Delta_r S°$$

and by $\Delta_r G° = -RT \ln K_{eq.}$

Consider the following reaction :

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

Given: $\Delta_f H°$ ($CH_3OH$, g) = –201 kJ/mol ;
$\Delta_f H°$ (CO, g) = – 114 kJ/mol
$S°(CH_3OH, g) = 240$ J/K-mol ;
$S°(H_2, g) = 29$ JK$^{-1}$ mol$^{-1}$
$S°(CO, g) = 198$ J/mol-K ;
$C°_{p,m}(H_2) = 28.8$ J/mol-K
$C°_{p,m}(CO) = 29.4$ J/mol-K ;
$C°_{p,m}(CH_3OH) = 44$ J/mol-K

and $\ln\left(\frac{320}{300}\right) = 0.06$, all data at 300 K

**Q22.** $\Delta_r S°$ at 300 K for the reaction is :
(A) 152.6 J/K-mol
(B) 181.6 J/K-mol
(C) –16 J/K-mol
(D) none of these

**Q23.** $\Delta_r H°$ at 300 K for the reaction is :
(A) –87 kJ/mol
(B) 87 kJ/mol
(C) –315 kJ/mol
(D) –288 kJ/mol

**Q24.** $\Delta_r S°$ at 320 K is :
(A) 155.18 J/mol-K
(B) 150.02 J/mol-K
(C) 172 J/mol-K
(D) none of these

**Q25.** $\Delta_r H°$ at 320 K is :
(A) –288.86 kJ/mol
(B) –289.1 kJ/mol
(C) –87.86 kJ/mol
(D) none of these

**Q26.** $\Delta_r G°$ at 320 K is :
(A) –48295.2 kJ/mol
(B) –240.85 kJ/mol
(C) 240.85 kJ/mol
(D) –81.91 kJ/mol

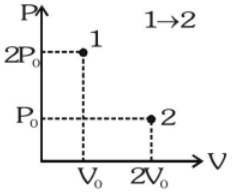
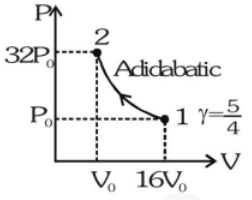
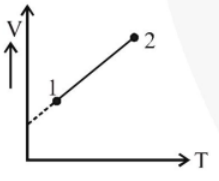
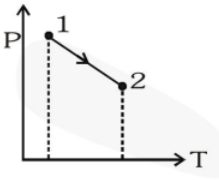
## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q27.**

| **Column-I** | **Column-II** |
|---|---|
| (A) Cyclic rule | (P) $\int -P_{ext.} \times dV$ |
| (B) Work done in reversible isothermal process | (Q) $\left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial T}{\partial P}\right)_V = -1$ |
| (C) Work done in isobaric process | (R) $-nRT\ell n\frac{P_1}{P_2}$ |
| (D) Work done during free expansion | (S) $-nRT\ell n\frac{V_1}{V_2}$ |
| (E) Work done in adiabatic reversible process | (T) $-nR(T_f - T_i)$ |
| (F) Work done in adiabatic irreversible process | (U) Zero |
| | (V) $\frac{nR}{\gamma - 1}(T_f - T_i)$ |
| | (W) $-nC_V \cdot T_{initial}\left[1 - \left(\frac{P_2}{P_1}\right)^{R/C_P}\right]$ |
| | (X) $-P_{ext.} \times nR\left[\frac{P_1T_2 - P_2T_1}{P_1P_2}\right]$ |

**Q28.**

| | **Column-I (For a definite amount of an ideal gas)** | **Column-II (Enthalpy change work done)** |
|---|---|---|
| (A) |   | (P) $\Delta H > 0$ |
| (B) |   | (Q) $\Delta H = 0$ |
| (C) |   | (R) $W > 0$ |
| (D) |   | (S) $W < 0$ |

**Q29.**

| **Column–I** | **Column-II** |
|---|---|
| (A) Reversible adiabatic compression | (P) $\Delta S_{system} > 0$ |
| (B) Reversible vaporisation | (Q) $\Delta S_{system} < 0$ |
| (C) Adiabatic free expansion of ideal gas in vacuum | (R) $\Delta S_{surroundig} < 0$ |
| (D) Dissociation of $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$ | (S) $\Delta S_{surrounding} = 0$ |

**Q30.** Match the following :

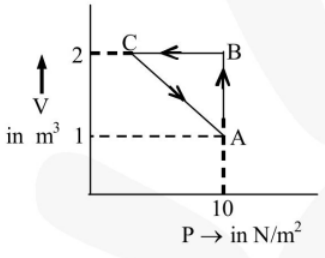
| **Column-I (Process)** | **Column-II** |
|---|---|
| (A) $\Delta H = q_p$ | (P) $\Delta H_f^0 = 0$ |
| (B) Kirchhoff's equation | (Q) A definite quantity |
| (C) $H^+(aq.)$ | (R) Path function |
| (D) Spontaneous process | (S) $\Delta G > 0$ |
| | (T) $\Delta S_{Total}^0 > 0$ |
| | (U) $\Delta H_2 - \Delta H_1 = \Delta C_P(T_2 - T_1)$ |

**NUMERICAL TYPE**

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

**Q31.** One mole of benzene liquid is converted into vapour at its boiling point (80°C). Find out change in internal energy ΔU for this process (in KJ).
(given → $\Delta_{vap.}H$ = 3.0 KJ/mol,
R = 8.314 J/k mole.)

**Q32.** An ideal gas is taken through the cycle A ⟶ B ⟶ C ⟶ A as shown in the figure. If net heat supplied to the gas in the cycle is 5J, the work done on the gas (in J) in the process C → A is -




**Q33.** Water expands when it freezes. Determine amount of work done in joules, when a system consisting of 1.0 L of liquid water freezes under a constant pressure of 1.0 atm and forms 1.1 L of ice.

**Q34.** The entropies of $H_2(g)$ and H(g) are 130.6 and 114.6 J $mol^{-1}$ $K^{-1}$ respectively at 298 K. Using the data given below calculate ΔH° **(in kJ/mol)** of the reaction given below.
$H_2(g) \longrightarrow 2H(g)$ ; ΔG° = 406.62 kJ/mol

**Q35.** One mol of non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K) to (4.0 atm, 5.0 L, 245 K) with a change in internal energy (ΔU) = 30.0 L-atm. The change in enthalpy (ΔH) of the process in L-atm.

**Q36.** The molar heat capacity of water at constant pressure, C, is 75$JK^{-1}$ $mol^{-1}$. When 1.0KJ of heat is supplied to 100g of water which is free to expand, the increase in temperature of water (in K) is :

**Q37.** The molar heat capacity of water is 75 $JK^{-1}$ $mole^{-1}$. What is the amount of heat required (in KJ) to raise the temperature of 100g of water from 300K to 302.4 K ?

**Q38.** Standard entropy of $X_2$, $Y_2$ and $XY_3$ are 60, 40 and $50 JK^{-1}mol^{-1}$, respectively. For the reaction, $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3 \quad \Delta H = -30 K.J$
to be at equlibrium the temperature (in K) will be:-

## Answer Key

| Ex-1 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| C | A | A | B | C | D | A | A | C | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | A | B | C | B | B | B | C | C | A |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| C | C | C | D | C | C | A | D | C | C |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| D | B | C | B | B | B | C | B | D | A |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| D | C | A | C | A | A | B | A | A | B |
| 51 | 52 | | | | | | | | |
| C | B | | | | | | | | |

## Ex-1A

**1.** B,C **2.** A, D **3.** A,B,C,D **4.** A,B,C,D **5.** A, B, D

**6.** A,B,D **7.** A,B,C **8.** C **9.** A,B,C,D **10.** B,C

**11.** C **12.** C,D **13.** B,C,D **14.** B,D **15.** A,C

**16.** A,C,D **17.** A,B **18.** A,C,D **19.** A,D

**20.** D **21.** B **22.** D **23.** B **24.** B

**25.** B **26.** B **27.** C **28.** A **29.** B

**30.** D **31.** A

**32.** A-R; B- S; C- P; D- Q

**33.** A→S, B→P, C→Q, D→R

**34.** 2 → 3 & 5 **35.** 2 **36.** 9 **37.** 2 **38.** A

**Ex.2**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| C | C | C | D | B | D | D | C | A | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | A | C | C | B | C | D | C | B | A |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| A | C | B | C | A | D | D | D | B | A |
| 31 | 32 | | | | | | | | |
| C | C | | | | | | | | |

## Ex-2A

**1.** A, B, C,D **2.** A,B, C, D **3.** A **4.** A, B, C,D **5.** A,C

**6.** B, C, D **7.** A, B, C **8.** D **9.** B,C,D **10.** B,C

**11.** A,B,C,D **12.** A,B,D **13.** B,C,D **14.** B

**15.** B,C **16.** B **17.** D **18.** C **19.** A

**20.** C **21.** D **22.** C **23.** A **24.** D

**25.** C **26.** D

**27.** A → Q ; B → P,R,S ; C → P,T ; D → U; E → P,V,W ; F → P,X

**28.** (A) → Q, S ; (B) → P, R ; (C) → P, S ; (D) → P, S

**29.** (A)→S ; (B) → P,R ; (C) →P, S (D) → P, R

**30.** (A) → Q ; (B) → U ; (C) → P ; (D) → T

**31.** 5.93 KJ **32.** 5.00 **33.** –10.13 J **34.** 436.00

**35.** 44.00 **36.** 2.50 **37.** 1.00 **38.** 750.00